EXECUTION EVIDENCE · TEAM CHAINBRAIN · INTERNATIONAL AI TRACK · TOPIC F · 20 OCT 2026

25 transactions on a public FISCO chain. 9 of them are attacks the card refused.

Every hash below opens in the hosted explorer, and one command re-fetches all 25 receipts from the public RPC. A refused payment is a success here: it broke the owner's rules, so the chain rejected it and no money moved.

25 of 25
receipts on the public chain match the expected result

16 + 9
16 allowed actions worked; 9 attempts outside the rules were refused

HK$0
moved by any of the 9 refused attempts

Check all 25 yourself in one command. Python 3, standard library only, read-only, no key needed:

```
python3 docs/submission/evidence/check_hosted.py
```

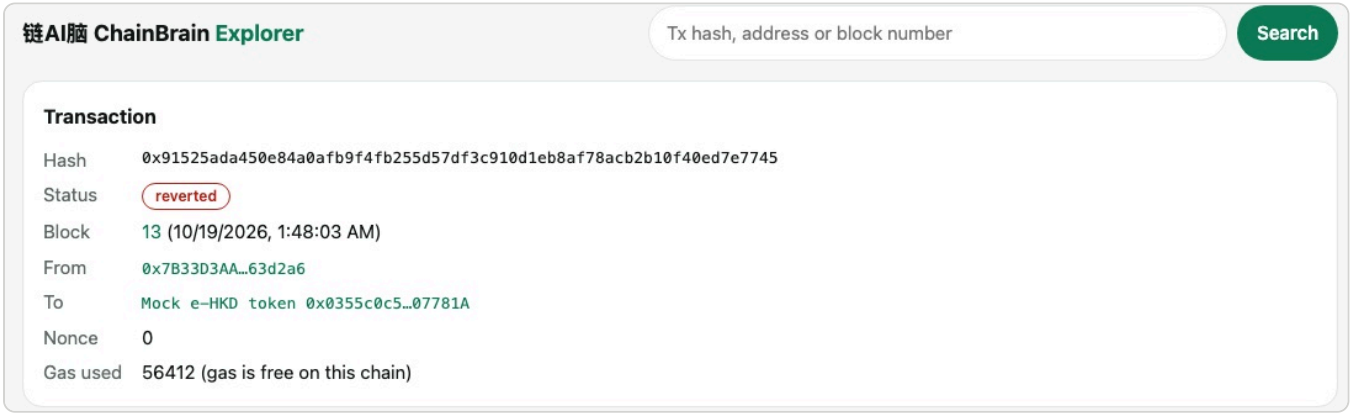
01 · EVIDENCE SUMMARY

Topic and scenario

Topic F, Agent Wallet. An owner gives an AI agent its own card: a smart contract holding Mock e-HKD and the owner's rules (which sellers, how much per payment, when to ask). The agent buys research reports from x402 sellers. Every payment is an EIP-3009 transfer from the card, which the token approves only after asking the card (ERC-1271).

What the evidence shows

- The agent's key pays an allowed seller alone, inside the rules.
- Above the ask-me limit, a payment settles only with the owner's co-signature.
- A weekly allowance is counted on chain; the owner's co-signature lifts it once.
- A compromised agent key sending straight to the chain is refused, every way we tried.



Transaction 5 in the hosted explorer: a compromised agent key sent a seller's planted transfer straight to the Mock e-HKD token, and the card refused it.

Not listed: test-suite and live-model runs, which ran on a local FISCO node judges can't reach. Those are in `docs/evidence-fisco.json`.

02 · TRANSACTIONS

Every hosted transaction

Time and block are read from the chain. "Refused" means the transaction was recorded on the chain and reverted, so nothing changed. The last column says which rule stopped it: for set B, the card's own check read just before each attempt; for set A, the case the attempt was built to test.

#	Time	Block	Action	Result	Why it was refused	Transaction
A · The app's own deployment · 19 October 2026, HKT						
1	01:47:52	9	Owner creates a research card: DataLite and DataPro, co-sign above the limit	done		0x9a1587e0…a958d9
2	01:47:53	10	Owner funds the card	done		0x84d74efe…cbb2ca
3	01:47:55	11	Agent pays DataLite over x402, agent key alone	done		0x15b0d30f…cf9333
4	01:47:56	12	Agent pays DataPro above the ask-me limit, owner co-signed	done		0xcc79bc34…6164a3
5	01:48:03	13	Compromised agent key sends a seller's planted transfer straight to the token	refused	payee not on the card's list	0x91525ada…7e7745
6	01:48:06	14	Owner creates an allowance card: DataLite only, never asks	done		0x28d40fb7…ff1afe
7	01:48:06	15	Owner funds it with HK$10	done		0x2247a643…f489ee
8	01:48:07	16	Owner sets an allowance of HK$3.00 a week	done		0x7a1cd383…7d5336
9	01:48:08	17	Agent pays HK$2.30, counted against the allowance on chain	done		0x7626c8aa…f1332c
10	01:48:09	18	Agent key alone, second report: over the week's allowance	refused	over the week's allowance	0xdf543023…392a9c
11	01:48:10	19	The same payment with the owner's co-signature	done		0xc0673643…808e93
B · Proof run on freshly deployed contracts · 19 October 2026, HKT						
12	18:02:35	28	Deploy a fresh Mock e-HKD	done		0xc17c58de…067f69
13	18:02:35	29	Deploy a fresh PocketFactory	done		0x8a3ba306…2a9098
14	18:02:36	30	Mint HK$50 to the owner	done		0x49222bc1…cdf0ce
15	18:02:36	31	Owner creates a card: cap HK$15, co-sign above HK$8, DataLite and DataPro	done		0x6d2c951f…f39053
16	18:02:37	32	Owner funds it with HK$40	done		0x0ff9192e…d173e4
17	18:02:37	33	Agent key pays DataLite HK$2.30	done		0xd687ae82…7bd2db
18	18:02:38	34	DataPro HK$11.70 with the owner's co-signature	done		0x8625dbfb…8d7386
19	18:02:38	35	Agent key sends HK$5 to an address a seller planted	refused	payee not on the card's list	0xd49be111…ed6954
20	18:02:39	36	Agent key pays DataPro HK$11.70 without the owner	refused	above HK$8, no owner signature	0xb3d88830…6277da
21	18:02:39	37	Agent key signs in the owner's place too	refused	owner signature from the wrong key	0xaf285aa5…506e87
22	18:02:40	38	Agent key pays DataLite HK$20, above the cap	refused	over the HK$15 cap	0xf0ad609b…99b82f
23	18:02:41	39	The HK$2.30 authorisation sent again (replay)	refused	already used once (replay)	0x84e1d74e…9c5384
24	18:02:41	40	Agent key tries to raise the cap to HK$800	refused	only the owner can change rules	0x7a3b37d1…9bfecf
25	18:02:42	41	Agent key tries to add the planted address as a seller	refused	only the owner can change rules	0x49c76450…5b9616

In rows 19 to 23 the agent's key signs and another key sends the transaction, as an attacker holding the agent key would; rows 24 and 25 are sent by the agent key itself. Set B's card ends with HK$26.00 = HK$40 − HK$2.30 − HK$11.70. Contract addresses are on page 3.

03 · REPRODUCTION GUIDE

Check it yourself

Nothing below needs a key or an account. Steps 1 to 3 only read the public chain.

1 · Open any transaction

Click a hash on page 2, or open `https://fisco-api-production-85c5.up.railway.app/explorer#/tx/<hash>`. The explorer shows status, block, sender, the contract called and its events.

2 · Re-fetch all 25 receipts

Python 3, standard library only. It reads the public RPC, prints block, time and status for every hash, and exits with an error if any result differs from the table.

```
python3 docs/submission/evidence/check_hosted.py
```

3 · Read a card's rules from the chain

The independent verifier rebuilds a card's rules in plain words and checks it was created by the ChainBrain factory. Add `--approved approved.json` to compare it with what the owner approved.

```
python3 verify/verify_pocket.py --rpc https://fisco-node-production.up.railway.app \
  --pocket 0x91019dDCD226aA00D3e76CD79C5bCFF9973D5374
```

4 · Run the proof again, with new keys

Deploys a fresh token, factory and card from new keys funded by the public faucet, then sends set B's 14 transactions. New hashes, same results; it exits with an error on any mismatch. Needs Node, `pnpm install` in `server/` and the contracts built with `forge build`.

```
cd server && OUT=/tmp/proof.json npx tsx scripts/hosted-proof.ts
```

5 · Try the app

Console: fisco-web-production.up.railway.app, mirror fintechathon-lian-ai-nao.vercel.app (Judges tab: four one-tap attacks). Test money: fisco-api-production-85c5.up.railway.app/faucet, HK$100 per address per day.

Chain details

Chain	FISCO BCOS 3.16.4, chain id 20200, one node hosted on Railway, gas price 0
Public RPC	`https://fisco-node-production.up.railway.app`
Explorer and faucet	`https://fisco-api-production-85c5.up.railway.app/explorer` · `/faucet`
Set A contracts	`token 0x0355c0c5Afe2A35369056FBB854903A4aF07781A` · factory `0x03aB4C311eeE7C3295c63F910D79a07Ffa7A097A` · allowance card `0x91019dDCD226aA00D3e76CD79C5bCFF9973D5374`
Set B contracts	`token 0x2299E8cD0be36226A43E1f7Ad7c4B367ABDFfF22` · card `0x33Dc83121F0D5A9cCfa0F1b55553B6Ee06013356`
Receipts file	`docs/submission/evidence/hosted-receipts.json`, written by step 2 with `--write`

Mock e-HKD is a test token on FISCO BCOS, not the HKMA's e-HKD.